DeepWeeds Image Classification for Targeted Herbicide Spraying

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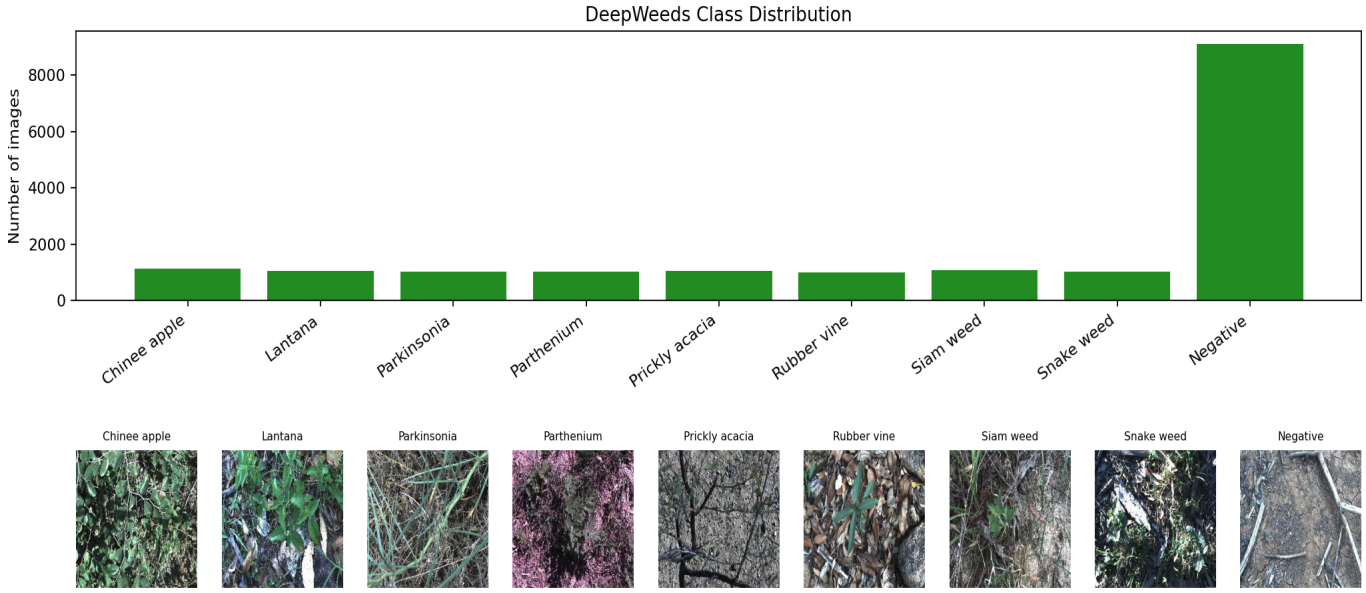
1. Introduction

Weeds compete with crops for essential resources. This project uses computer vision to classify field images into eight weed species and a Negative class, which represents images without the target weed. Because class imbalance can make accuracy misleading, macro F1 and per-class metrics are also reported.

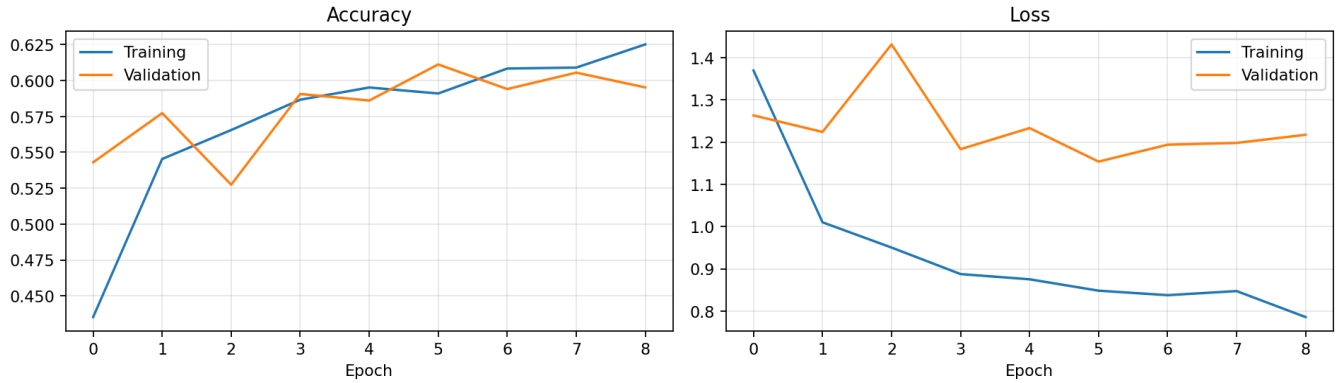
2. Dataset and Methodology

The DeepWeeds dataset contains 17,509 images. Stratified splitting with random seed 42 produced 12,256 training, 2,626 validation, and 2,627 test images. MobileNetV2 pretrained on ImageNet was used as the CNN backbone. Images were resized to 160 × 160 pixels, augmented using horizontal flipping, rotation and zoom, and trained with class weights. A global average pooling layer, dropout and nine-class softmax classifier formed the classification head.

Dataset class counts and representative samples



Training and validation learning curves

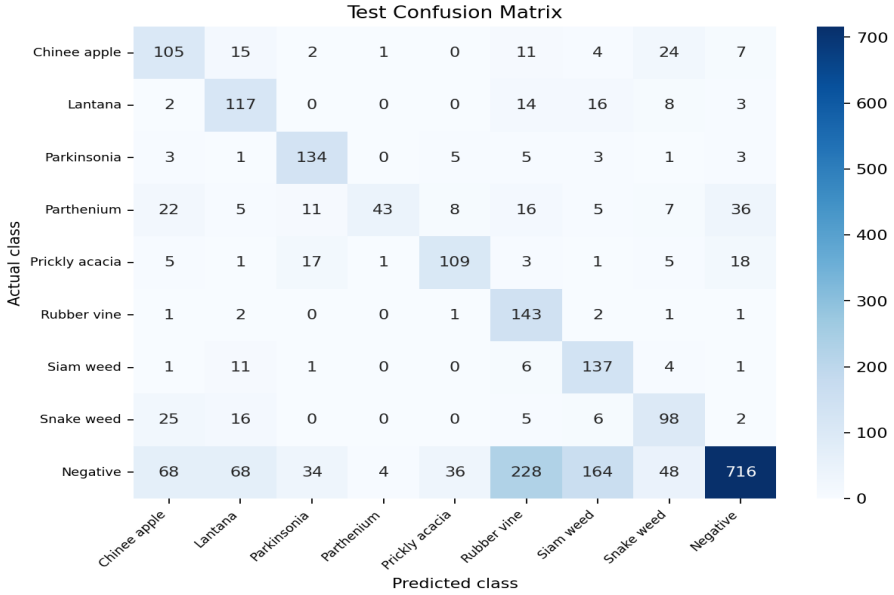


3. Results and Discussion

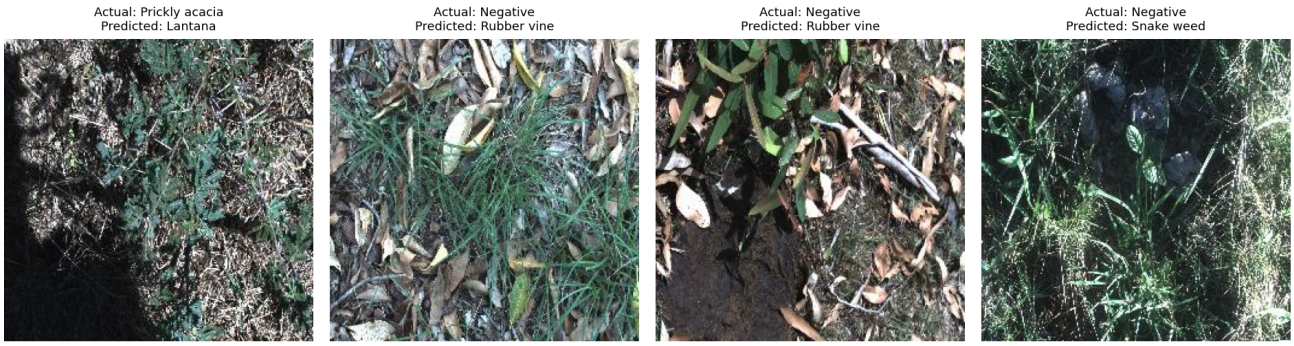
Test accuracy: **60.98%** Macro F1-score: **0.5833** Test images: **2,627**. Macro F1 gives equal weight to each class, helping reveal uneven performance that overall accuracy may hide.

Class	Precision	Recall	F1-score	Support
Chinee apple	0.453	0.621	0.524	169
Lantana	0.496	0.731	0.591	160
Parkinsonia	0.673	0.865	0.757	155
Parthenium	0.878	0.281	0.426	153
Prickly acacia	0.686	0.681	0.683	160
Rubber vine	0.332	0.947	0.491	151
Siam weed	0.405	0.851	0.549	161
Snake weed	0.500	0.645	0.563	152
Negative	0.910	0.524	0.665	1366

Confusion Matrix



Four Misclassified Test Images



4. Conclusion

MobileNetV2 achieved the reported test performance, providing a baseline for weed recognition. The confusion matrix, per-class scores and errors should guide further improvements, such as fine-tuning, better augmentation and collecting more examples for difficult classes. The model requires further field validation before use in automated spraying, where false positives can waste herbicide and false negatives can leave weeds untreated.

References: Olsen et al., DeepWeeds, Scientific Reports (2019), https://doi.org/10.1038/s41598-018-38343-3; TensorFlow MobileNetV2 documentation; https://github.com/AlexOlsen/DeepWeeds.